
Advanced Computer Graphic Report

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1 Introduction

This project implements physically based lighting effects and supports diffuse materials, perfect mirror materials, glass materials, and metallic materials. Rendering performance is improved through multithreading and BVH acceleration, while Intel Open Image Denoise is integrated for image denoising. The project also supports basic environment lighting and uses multiple importance sampling to reduce noise. In addition, Instant Radiosity and Light Tracing are introduced as additional rendering techniques.

2 Extensions to the Path Tracer

2.1 Rough Conductors

The project implements a rough metallic material based on the GGX microfacet model. During sampling, the outgoing direction is transformed into the local shading frame, and two random samples are used to generate a microfacet half-vector, from which the incident direction is computed by ideal reflection. During evaluation, the half-vector is reconstructed to compute the conductor Fresnel term, GGX normal distribution term, and Smith geometry term. These components are combined with the surface albedo to produce the reflected contribution, and the corresponding PDF is derived from the same half-vector.

The microfacet BRDF is

$$f_r(x, \omega_i, \omega_o) = \frac{G(\omega_o, \omega_i) D(\omega_m) F(\omega_o)}{4 \cos(\theta_o) \cos(\theta_i)}.$$

For GGX, the normal distribution function is

$$D(\omega_m) = \frac{\alpha^2}{\pi} (\cos^2 \theta_m (\alpha^2 - 1) + 1)^2,$$

The corresponding PDF for half-vector sampling becomes

$$p(\omega_i) = \frac{D(\omega_m) \cos(\theta_m)}{4(\omega_o \cdot \omega_m)}.$$

2.2 Glass

The outgoing direction is first transformed into the local shading frame, and the sign of its cosine with the surface normal is used to determine whether the ray is entering or leaving the medium, which defines the refractive index relationship. The Fresnel term is then used to compute the probability of reflection, while Snell's law is used to determine whether a valid refracted direction exists. If total internal reflection occurs, only ideal reflection is produced; otherwise, the interaction is probabilistically chosen between ideal reflection and ideal refraction according to the Fresnel reflectance. Since this material is treated as a delta distribution, it only contributes along the ideal reflected or refracted direction, and therefore BRDF evaluation and PDF queries for ordinary directions both return zero.

For dielectric materials, the Fresnel term is computed from the parallel and perpendicular components:

$$F^{\parallel}(\omega) = \frac{A_r^{\parallel}}{A_r^{\parallel}} = \frac{\cos(\theta_i) - \eta \cos(\theta_t)}{\cos(\theta_i) + \eta \cos(\theta_t)}$$

$$F^{\perp}(\omega) = \frac{A_r^{\perp}}{A_r^{\perp}} = \frac{\eta \cos(\theta_i) - \cos(\theta_t)}{\eta \cos(\theta_i) + \cos(\theta_t)}$$

and the final dielectric Fresnel reflectance is

$$F(\omega) = \frac{F^{\parallel}(\omega)^2 + F^{\perp}(\omega)^2}{2}.$$

The transmitted direction is constrained by Snell's law:

$$\eta_i \sin(\theta_i) = \eta_t \sin(\theta_t).$$

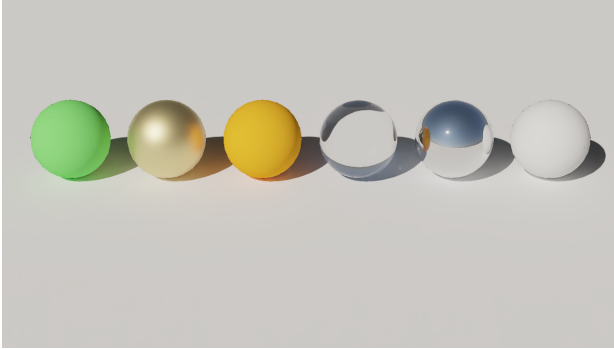


Figure 1: material

2.3 BVH Acceleration and Tile-Based Multithreaded Rendering

The BVH accelerates rendering by hierarchically enclosing geometry in bounding volumes, allowing the renderer to quickly discard large numbers of triangles that cannot be intersected by a ray. This significantly reduces ray-geometry intersection cost.

Tile-based multithreading improves CPU utilisation by dividing the image into smaller regions and letting multiple threads render these regions in parallel. This reduces the rendering time per sample and improves overall performance.

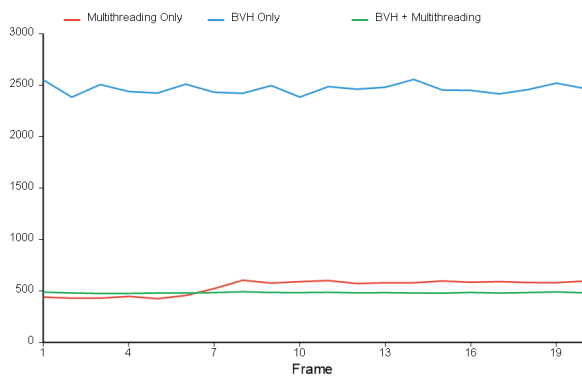


Figure 2: Rendering Time Comparison Under Different Acceleration Modes

2.4 Environment Lighting

Environment lighting is implemented using an environment map to represent incoming radiance from all

directions. During initialisation, a cumulative distribution is built from pixel luminance with a sine-based weighting term, so that bright regions are sampled more frequently while spherical area distortion is compensated for. During rendering, a sampled pixel is converted into a 3D direction, and the corresponding radiance is obtained from the environment map. Its PDF is computed from the pixel sampling probability and converted into a continuous directional density using the associated solid angle. When combined with BSDF sampling, multiple importance sampling is applied using the power heuristic. If no valid luminance distribution is available, the method falls back to uniform sphere sampling.

Power heuristic for multiple importance sampling:

$$w_j = \frac{p_j^\beta}{\sum_{k=1}^M p_k^\beta}$$

2.5 OIDN Denoising

In this project, the Film class stores aligned beauty, albedo, and normal buffers. During rendering, albedo and normal information are recorded for each visible surface sample, and these buffers are then converted into contiguous Float3 arrays and passed to Intel Open Image Denoise. The denoised result is finally written out as a PNG image.

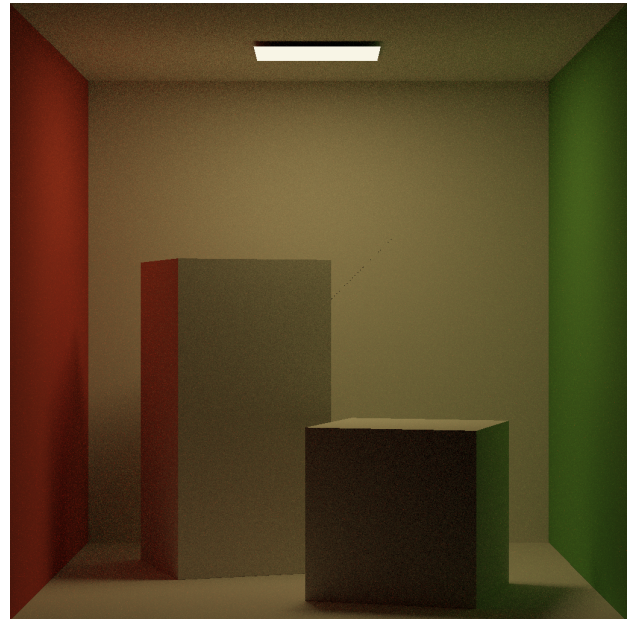


Figure 3: cornell-box without denoise



Figure 4: cornell-box with denoise

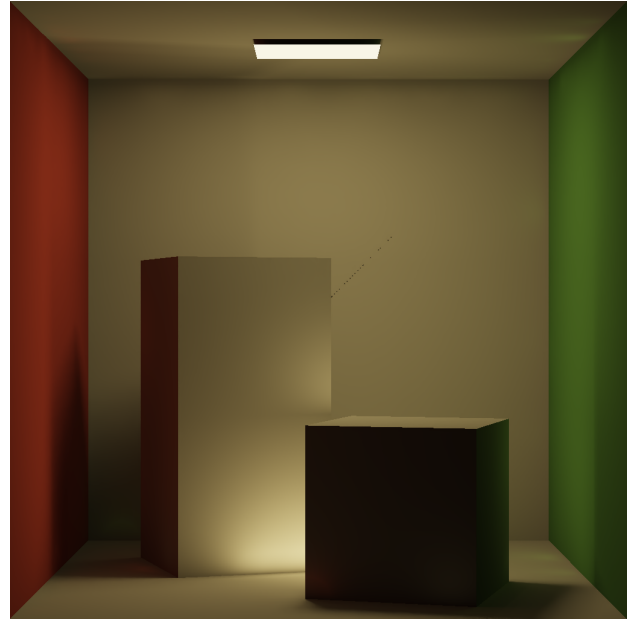


Figure 5: Instant Radiosity

3 Additional Rendering Technique

3.1 Instant Radiosity

Instant Radiosity approximates indirect illumination by first tracing light paths from the real light sources and storing non-specular interaction points along those paths as virtual point lights. At camera-visible surfaces, these VPLs are then treated as secondary light sources and connected to the visible point to estimate indirect lighting.

Each stored VPL keeps both its shading data and the throughput transported from the real light source. In `instantRadiosity()`, the renderer first handles any specular bounces on the camera path, then computes the ordinary direct lighting term, and finally iterates over all VPLs.

Using this, the indirect illumination estimate can be written as

$$L_o(x \rightarrow x_{camera}) = \sum_{i=1}^{N_{VPL}} f_r(x_{prev} \rightarrow x_i \rightarrow x) G(x_i \leftrightarrow x) f_r(x_i \rightarrow x \rightarrow x_{camera}) L_{VPL}(i)$$

Compared to path tracing, Instant Radiosity reduces noise, but achieving better results requires a large number of VPLs, which significantly increases rendering time and introduces additional lighting artifacts.

3.2 Light Tracing

This project implements a basic light tracing method. For each path, the renderer samples a light source, an emission position, and an outgoing direction to initialise the path throughput. The path is then propagated through the scene for a limited number of bounces. At each non-light intersection, the current vertex is connected to the camera by projection and visibility testing. If the connection is valid, the BSDF and cosine term in the camera direction are evaluated and the contribution is accumulated to the corresponding pixel. The next direction is then sampled from the BSDF, and the path throughput is updated accordingly.

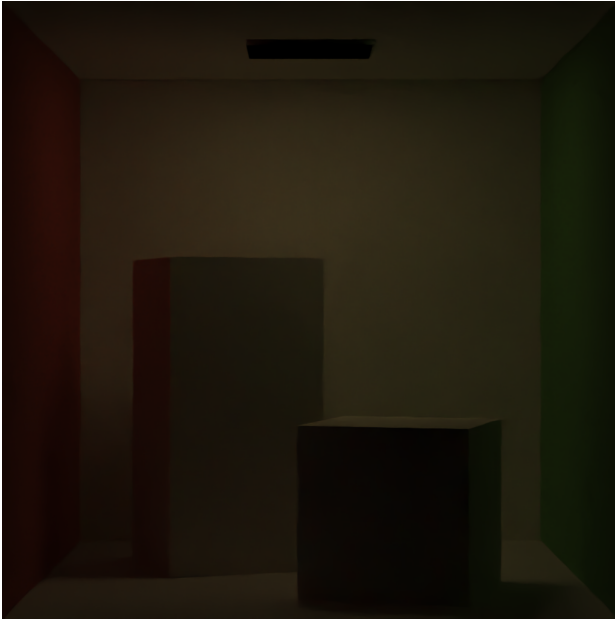


Figure 6: *light tracing*

Repository

Project repository: <https://github.com/AODINGLVP/WM9M3---.git>

Due to the lack of optimization, light tracing rarely results in paths that reach the camera, leading to an overall darker image and increased noise.

4 Reflection

If I were to complete this project again, I would add comments in the code during the early implementation stage to clearly indicate the formulas being used, so that later maintenance would be more convenient. At the same time, when adding BVH acceleration and multithreaded rendering, I would also implement switching functionality to make performance testing more convenient and efficient.

5 Conclusion

This project builds a physically based path tracing framework and integrates several core rendering techniques, including diffuse, perfect mirror, glass, and rough metallic material models, as well as environment lighting combined with multiple importance sampling. These techniques improve the physical realism of the rendered results.

By combining BVH acceleration with tile-based multithreaded rendering, the system significantly reduces ray intersection cost and overall rendering time, thereby improving performance. In addition, OIDN denoising is integrated to effectively reduce noise while preserving important details.

Furthermore, Instant Radiosity and Light Tracing are also introduced and compared. Overall, the system achieves a good balance between rendering performance and image quality.